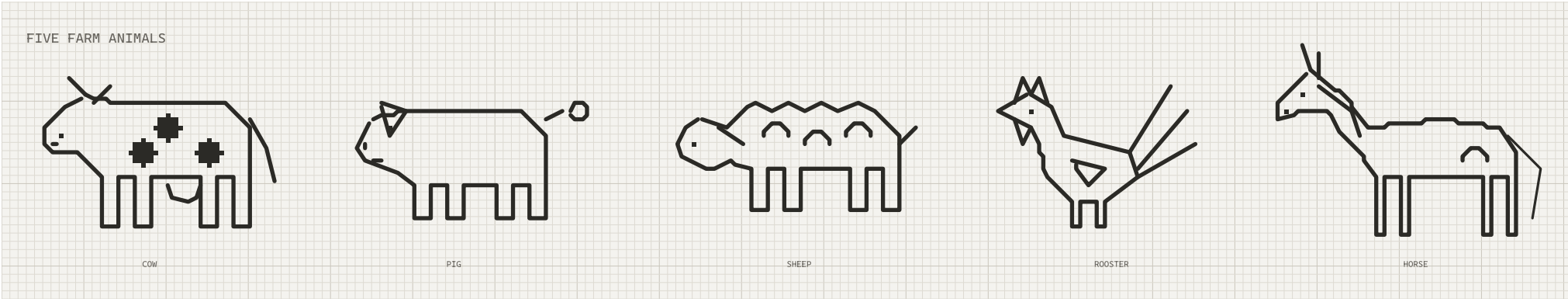


Five Farm Animals

Five animals drawn on TurtlePen’s integer lattice — every coordinate a whole 5 px quadrant, every defect a ranked finding. This is the complete record: how each silhouette is built, which findings were real defects and which were declared anatomy, what the exercise taught, the reordered `turtlepen_help` text, and the diagrams the project has produced so far.



200 × 36 cells · 62 elements · validate CLEAN: 0 critical, 0 error, 0 warn, 0 info · 11 findings accepted as intentional

01	02	03	04	05	06
How the animals are built	The five, one at a time	What the collision log caught	Things learned	The help was reordered	Every SVG so far

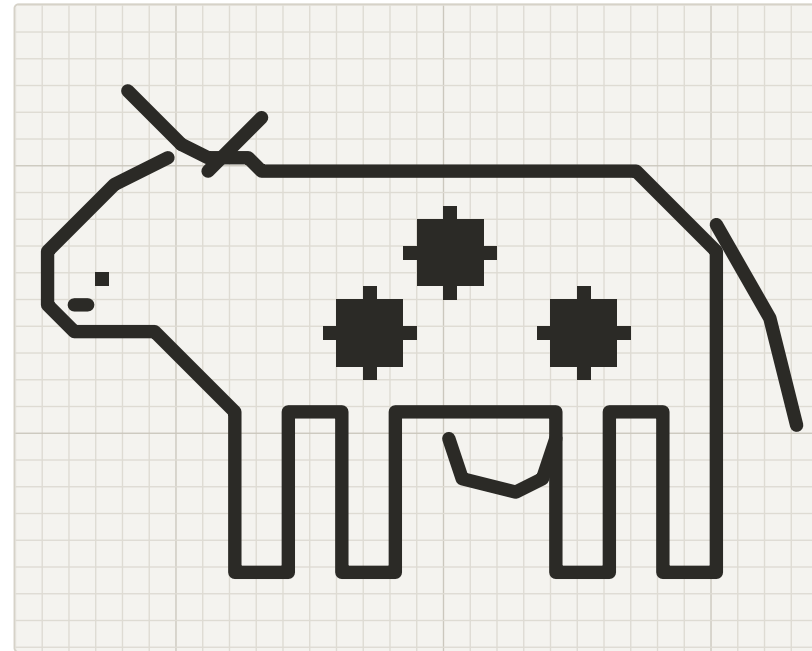
HOW THE ANIMALS ARE BUILT

One polygon, legs included. The obvious construction — a body outline plus four vertical leg strokes — fails twice. Each leg T-junctions into the belly and raises an **L006** stroke overlap, and starting the leg one row lower to dodge it leaves a dangling end instead. Folding the legs into the closed outline solves both, and is how a silhouette is genuinely drawn.

Reading the cow's back-to-front path: over the back, down the rump, then *down-across-up* for each leg with the belly between them, up the chest, round the head, and closed.

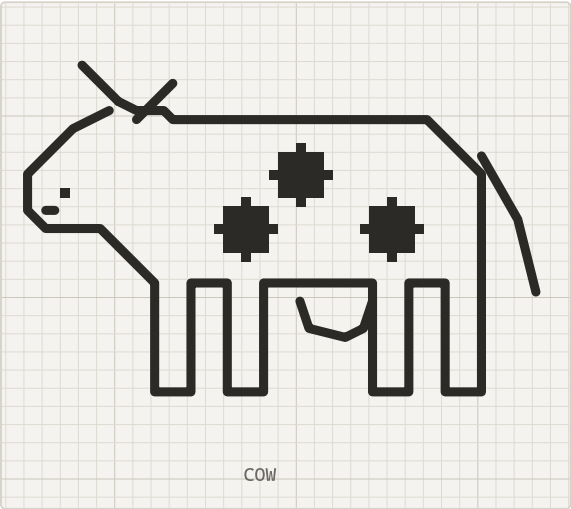
AE20 → AE26	hind leg 1, outer edge down
AE26 → AC26	the hoof
AC26 → AC20	inner edge back up
AC20 → AA20	belly between the legs
AA20 → AA26	hind leg 2 ... and so on

Everything else is a separate mark. Horns, ears, eyes, spots, udders, combs, wattles, manes and tails are their own paths with **role: "artwork"**, so they are not checked for loose ends. Where one touches the body — an ear meeting a skull — the shared quadrant is real, and is declared with **accept_finding** rather than dodged.



The cow at 1.4x. Legs, belly, chest, head and back are one continuous 25-point closed path; the horn, ear, eye, mouth, udder, three spots and tail are eight more.

THE FIVE, ONE AT A TIME



Cow

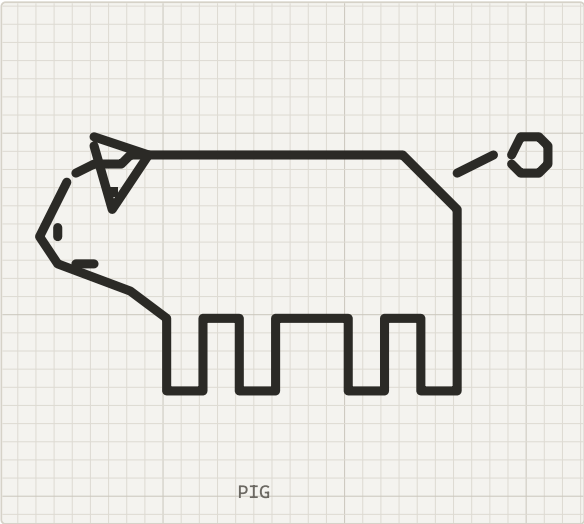
Head, neck, barrel and all four legs are a single 25-point closed polygon. The blunt muzzle is two points (F16 → F14) rather than one — a single apex read as a goat.

horn · ear · eye · mouth · udder · 3 solid spots · tail

```

polygon K10 O11 AB11 AE14 AE20 AE26 AC26 AC20 AA20 AA26
Y26 Y20 S20 S26 Q26 Q20 Q20 O26 M26 M20
J17 G17 F16 F14 H12

```



Pig

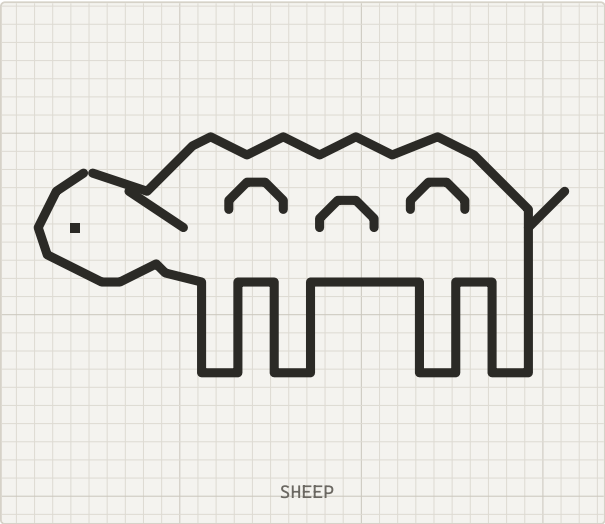
Same skeleton, different proportions: deeper barrel, legs stopped four rows short, and a blunt two-point snout. The ear is a triangle that deliberately crosses the skull line — that is how a pig's ear sits.

floppy ear · eye · snout crease · mouth · curly tail (ray + circle)

```

polygon AT13 AX12 BL12 B015 B021 B025 BM25 BM21 BK21 BK25
BI25 BI21 BE21 BE25 BC25 BC21 BA21 BA25 AY25 AY21
AV19 AS18 AR16 AS14

```



Sheep

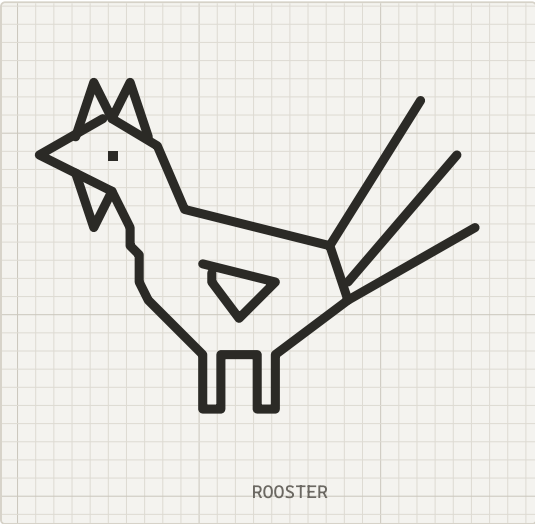
Fleece is the outline itself: ten alternating points across the back. The first attempt used 2-row spikes every 3 cells and read as a stegosaurus; 1-row bumps every 2 cells read as wool.

back-swept ear · eye · 3 interior fleece arcs · short upturned tail

```

polygon CH13 CK14 CM12 C011 CQ12 CS11 CU12 CW11 CY12 DA11
DC12 DD13 DF15 DF19 DF24 DD24 DD19 DB19 DB24 CZ24
CZ19 CT19 CT24 CR24 CR19 CP19 CP24 CN24 CN19 CK18
CI19 CF18 CE16 CF14

```



Rooster

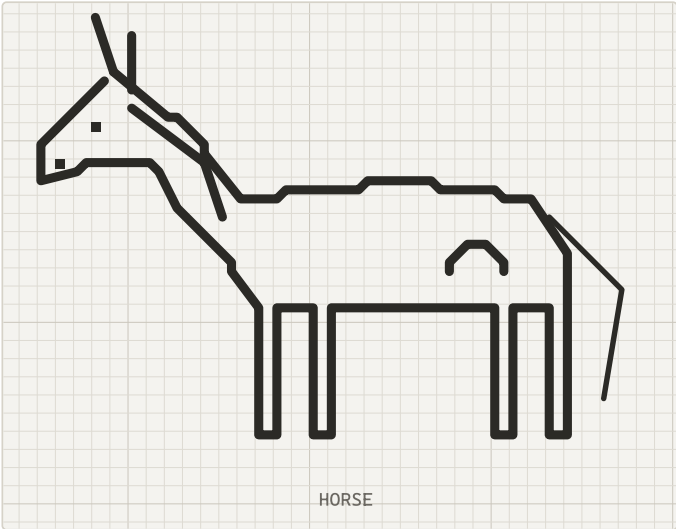
The only upright body. Two 1-cell legs instead of four 2-cell ones, and the tail is three separate rays fanning off the rump vertex rather than part of the outline.

zigzag comb · wattle · eye · triangle wing · 3 tail plumes

```

polygon DV10 DX11 DZ15 ED16 EH17 EI20 EE23 EE26 ED26 ED23
      EB23 EB26 EA26 EA23 DX20 DW16 DV14 DT13 DR12 DT11

```



Horse

Long legs and an arched neck built from an 8-point crest. The mane began as three short strokes across the neck and read as stitches; one line running parallel to the crest reads as a mane.

two separate ears · eye · nostril · mane line · haunch arc · 3px tail

```

polygon FD7 FH10 FK14 FT13 GA14 GC17 GC20 GC27 GB27 GB20
      FZ20 FZ27 FY27 FY20 FP20 FP27 F027 F020 FM20 FM27
      FL27 FL20 FJ17 FH15 FF12 FC12 EZ13 EZ11

```

WHAT THE COLLISION LOG CAUGHT

Real defects – fixed

RULE	DEFECT AND FIX
L006	horse-haunch crossed the rump outline The haunch arc (r=4 at GA18) reached col 185, exactly where the rump edge and hind-leg line sit. Moved to FX18 with r=3, fully inside the barrel.
L006	two horse tail strands shared 5 quadrants Both strands swept the same corridor. Three re-routes each collided somewhere else; the honest fix was one strand drawn at width 3 — a horse has one tail.
L006	pig ear covered the pig's eye Widening the ear to read as a flop put its inner edge through AV14. Caught in rehearsal, before commit: the ear was narrowed instead of moving the eye.
visual	cow spots read as plus-signs tone:"half" discs at r=2 dither to a 5-quadrant cross. Spots are solid — dropped the tone and went to r=3.
visual	cow and horse 'ears' read as flags A two-ray polyline from one root point renders as a filled-looking V. Split into separate single-ray marks.
visual	sheep read as a stegosaurus Amplitude, not concept: the fleece zigzag was too coarse. Halved the amplitude and doubled the frequency.

Declared anatomy – accepted

Each carries a written reason and lapses automatically if the geometry moves.

PATHS	QD	REASON
cow-body / cow-horn	1	horn grows from the poll vertex
cow-body / cow-ear	1	ear is rooted on the neckline behind the poll
cow-body / cow-udder	2	udder attaches to the belly line — it must touch
pig-body / pig-ear	3	a pig's ear flops forward across the skull line
sheep-body / sheep-tail	1	tail springs from the rump edge
rooster-body / rooster-comb	3	comb sits on the skull line, touching at three peaks
rooster-body / rooster-wattle	3	wattle hangs from the chin line at both ends
rooster-body / rooster-plume-1	1	plume grows out of the tail-base vertex
rooster-body / rooster-plume-3	1	plume grows out of the rump vertex
horse-body / horse-ear-1	1	near ear is rooted at the poll vertex
horse-body / horse-ear-2	1	off ear is rooted on the crest behind the poll

Five more lapsed. Redrawing the cow, sheep, horse and pig retired their earlier acceptances as *stale* rather than carrying them silently forward.

THINGS LEARNED

A clean log is not a finished drawing

`validate` returned CLEAN while the sheep read as a stegosaurus, the cow's horn and ear rendered as a flag, and the udder read as a smile. Every one of those was found by rasterising the SVG and looking at it. The help says this outright — it is now on page one of the help rather than line 212.

Draw legs as part of the outline, not as separate strokes

The obvious approach — a body polygon plus four vertical leg strokes — T-junctions every leg into the belly and raises an L006 per leg. Folding the legs into the closed outline gives exact geometry *and* better line art: it is how a silhouette is actually drawn. All five animals use it.

ascii proves structure; only a raster proves identity

The ASCII dump confirmed the cow's belly, leg gaps and spot placement were quadrant-exact. It could not tell me the head read as a goat. Use ASCII to verify you drew what you meant, and the render to verify it means what you wanted.

accept_finding is where anatomy gets declared

An ear touching a skull *is* a shared quadrant. Eleven findings were accepted with a stated reason; two that looked identical were real defects. Bulk-accepting L006 would have shipped both.

Lapsed acceptances did exactly their job

Redrawing the cow, sheep, horse and pig retired five acceptances as *stale* rather than carrying them forward. An acceptance that survived a geometry change would be indistinguishable from a missed defect.

Rehearse before commit – fixes introduce collisions

Widening the pig's ear put it through the pig's eye. `plan` without `commit` surfaced that against a throwaway copy, so the document never held the broken version.

Column arithmetic past Z is the main source of address error

Cols 27–190 span AA–GH. The ASCII header row is ground truth for what actually landed — I mis-read its column offset once and briefly believed three correct spots were misplaced.

A running MCP server holds its help in memory

Editing `src/mcp/tools.js` does not change what the connected session returns; `HELP` is captured at module load. The reorder was verified by importing the module directly, and reaches a session on the next server start.

Rasterising an SVG needed a fallback

The Playwright MCP backend timed out and stayed wedged, and `file:` URLs are blocked. Headless Chrome `--screenshot` against a local static server did the job in one call — and printed this PDF.

THE HELP WAS REORDERED

What moved. THE CANVAS IS NOT A BUDGET and WORKFLOW sat at help lines 197–231, after the whole grammar reference and the connector notes. They are now lines 9–44, immediately after WHY IT EXISTS and before THE LATTICE.

Why it matters. Those two sections carry the instructions an author most needs *before* drawing: that the canvas is not a budget, that a feature may be more than one stroke, that layers exist — and that the loop ends in *render, then look*. This session is the case in point: the log was clean while three of five animals were wrong. Reference material can be read on demand; the working method cannot be read after the fact.

Mechanics. 36 lines moved verbatim inside the HELP template literal in `src/mcp/tools.js`, verified content-identical by comparing the sorted line multiset before and after. `pnpm run check: 318 passed, 0 failed`. The help assertion in `test/mcp.test.js` checks for the presence of section names, not their order, so the move is safe by design.

One caveat. HELP is captured when the module loads, so an MCP session already connected keeps serving the old order until the server restarts.

Before

- 1. WHY IT EXISTS
- 2. THE LATTICE
- 3. ADDRESSING
- 4. FREE SPACE
- 5. REGIONAL DESCRIPTION
- 6. DIMENSIONED COMPOSITIONS
- 7. HISTORY AND RECOVERY
- 8. GROUPS AND FOLLOW RELATIONSHIPS
- 9. PEN GRAMMAR
- 10. SHAPES
- 11. ANCHORS
- 12. ARTWORK PRESENTATION
- 13. TONE
- 14. DRAWING FROM A SOURCE
- 15. EXAMPLE
- 16. CONNECTORS
- 17. THE CANVAS IS NOT A BUDGET
- 18. WORKFLOW
- 19. EVERY FIX HAS A TOOL

After

- 1. WHY IT EXISTS
- 2. THE CANVAS IS NOT A BUDGET
- 3. WORKFLOW
- 4. THE LATTICE
- 5. ADDRESSING
- 6. FREE SPACE
- 7. REGIONAL DESCRIPTION
- 8. DIMENSIONED COMPOSITIONS
- 9. HISTORY AND RECOVERY
- 10. GROUPS AND FOLLOW RELATIONSHIPS
- 11. PEN GRAMMAR
- 12. SHAPES
- 13. ANCHORS
- 14. ARTWORK PRESENTATION
- 15. TONE
- 16. DRAWING FROM A SOURCE
- 17. EXAMPLE
- 18. CONNECTORS
- 19. EVERY FIX HAS A TOOL

Verbatim, as the tool returns it after the reorder. Also shipped beside this PDF as `turtlepen-help.txt`.

TurtlePen – an integer-exact grid for AI-authored diagrams.

WHY IT EXISTS

Diagram tools measure text at render time, long after a layout was chosen, so labels overflow their boxes and nothing notices. Here measurement happens before placement, every coordinate is an integer, and every defect is reported with a severity and a numeric fix. Nothing is ever silently resized.

THE CANVAS IS NOT A BUDGET

The grid is unbounded right and down. 160x100 is a starting size, not a limit, and `set_canvas` grows it. If a shape is cramped, MAKE IT BIGGER – an author who fights for room inside a size they picked early has mistaken their own first guess for a constraint.

Two more things that are easy to forget you have:

- A feature can be MORE THAN ONE STROKE. If detail would damage a shape by being carved out of it, draw a second mark beside it instead. Additive beats subtractive: subtracting from a stroke that carries the meaning destroys the thing you are annotating.
- Layers. `add_page` with intent "overlay" puts marks ON TOP without an L001, so annotation, texture, and construction can live apart from the artwork instead of competing with it for the same quadrants.

WORKFLOW

measure -> plan -> commit -> validate -> render -> LOOK AT IT
-> accept_finding for anything deliberate.

Only a fingerprint in the current validation may be accepted. Geometry changes make the old acceptance visibly stale; `unaccept_finding` withdraws that record. A committed edit that proves wrong is recoverable with `history action="undo"`.

Rendering and looking is part of the loop, not an optional last step. A clean log is evidence the drawing is undefective, never that it is finished – a corpus once validated clean while a rug sat 60 cells from the sofa and an apple's stem floated clear of the fruit. Use "ascii" to read the actual quadrants; it is cheaper than a render and catches a malformed shape fastest.

"plan" is the important one: send the whole composition as a batch of operations and read the collision log BEFORE anything is committed. The document is untouched until you re-send with `commit:true`, and a batch that fails part-way applies nothing at all.

Findings are ranked S0 critical, S1 error, S2 warn, S3 info. Accepting a finding records intent; it lapses automatically if the geometry changes.

THE LATTICE

1 cell = 10x10 px. 1 quadrant = 5x5 px. Strokes are 5px = 1 quadrant thick. Every legal position is a whole number of quadrants, so results are exact.

ADDRESSING (Excel)

C4 whole cell C4.tl pin point (tl t tr l c r bl b br)
C4.q2 quadrant (q1..q4) origin A1 top-left, unbounded right and down
Start at an inset origin such as T20 if the drawing may grow up or left.

FREE SPACE

`free_space` defaults to `scope="stack"`: all non-reference pages constrain the

answer, including hidden pages because validation still checks them. Every response names `searched_pages`. Use `scope="page"` only when cross-page overlap is intentional. Supply `cellsW` and `cellsH` together, or neither to list regions.

REGIONAL DESCRIPTION

`describe { region: "C4:AZ40" }` returns only elements whose exact claimed rectangles touch that range. Paths are tested piece by piece, not by a loose bounding box. Add page to narrow both page and region; the response repeats the normalized effective filter.

DIMENSIONED COMPOSITIONS

`wireframe` authors a plan/elevation in real inches and measures routed runs; `export_prompt` turns that stored composition into a normalized image brief. The source survives save/open. Before export, TurtlePen verifies every generated box and route still matches it; stale geometry is refused by name. Undo the manual edit or rerun `wireframe` rather than exporting old facts. `perspective_scene` projects real room inches through a declared camera and preserves those inputs as provenance with the document.

HISTORY AND RECOVERY

<code>history { action: "status" }</code>	inspect the recovery boundary
<code>history { action: "undo" }</code>	reverse the newest mutation
<code>history { action: "redo" }</code>	reapply the newest undone edit
<code>history { action: "clear" }</code>	discard undo and redo entries

The newest 100 successful document mutations are recoverable by default (`TURTLEPEN_HISTORY_LIMIT` accepts 1..1000). Failed and no-op calls do not consume history; a new edit after undo clears redo. A versioned sidecar is bound to the exact document hash, so undo/redo survives open and process restart. Outside edits invalidate stale history rather than replaying it against the wrong file.

GROUPS AND FOLLOW RELATIONSHIPS

<code>group { action: "create", id, members }</code>	own a flat subsystem
<code>group { action: "move", id, cellsX, cellsY }</code>	move every member exactly once
<code>constraint { action: "create", id, dependent, target }</code>	keep current anchor relationship
<code>constraint { action: "sync", id }</code>	restore the stored offset

An element belongs to at most one group. A dependent has at most one parent; chains cascade and cycles are refused. Named and indexed anchors work, offsets are whole quadrants, and describe reports groups plus relationship sync state.

PEN GRAMMAR

<code>pen from <id>.<N S E W></code>	START HERE for connectors: seats the cursor just outside a box's face, already facing away from it
<code>pen from <id>.<face>#<slot></code>	fan out competing connectors; #1 is the midpoint, #2 left/up, #3 right/down, then alternate by one cell (example: gateway.S#2)
<code>pen <address> [<pin>]</code>	place the cursor by address
<code>face <dir></code>	turn without drawing
<code><dir> [n] [align <side>] [<style>] line</code>	draw n cells of 5px stroke
<code><dir> [<style>] corner align <sideA> <sideB></code>	place a junction and turn
<code><dir> ... line to <address id.port></code>	draw until it reaches a target;

indexed target faces also work

SHAPES – anything that is not a rectangle

ray to <address>	a straight line at ANY angle
circle <r>	outline; radius in quadrants
disc <r>	the same circle, filled
arc <r> <startDeg> <endDeg>	clockwise from east
polygon <addr> <addr> <addr> ...	closed
triangle <addr> <addr> <addr>	exactly three points
dot [<dir8>]	one quadrant
dash <n> <dir8>	n quadrants, any of eight ways
dir8 = n ne e se s sw w nw	(up/down/left/right also accepted)

ANCHORS – position as a relationship, not a coordinate

pen at <id>.<anchor>	put the cursor ON an element
<shape> ... at <id>.<anchor>	anchor a shape to one
<shape> ... at <id>.<anchor> offset <dx> <dy>	nudge in whole quadrants

anchors: N NE E SE S SW W NW C

"from" gives the SEAT, one step OUTSIDE the element, where a connector starts.
"at" gives the anchor itself, on the element, where a shape belongs. Anchoring avoids hand-computed placement drift when the program runs. This grammar alone does not store a relationship: rerun the declarative program to recompute it, or create an explicit constraint when the dependent must follow later edits.

These are integer algorithms – Bresenham for lines, midpoint for circles – so the same command always covers the same quadrants. A stepped diagonal is not an approximation of a line; on a lattice it IS the line.

<dir> arrow	arrowhead pointing that way
<dir> hop	deliberate crossing (exempt from L006)

box span <W>x<H> at <address> label "... " [style <s>] [id <name>]
text "... " at <address> [span <W>x<H>] [id <name>]
[font <px>] [fill <#hex>] [weight <100..900>] [align left|center|right]

ARTWORK PRESENTATION (arguments on the pen tool or plan operation)

role: "artwork"	open marks are not connectors
color: "#rrggbb", width: 1..5, cap: "round"	continuous presentation ink
paint: "cells"	colour every exact claimed quadrant

TONE – density, and why it is not opacity

tone: 0.0625..1	or "quarter" "half" "three-quarter" "solid"
feather: <n>	quadrants of falloff inward from the region boundary
texture: "eroded"	seeded rough edge, deterministic

tone changes WHAT IS INKED. A half-tone shape inks half its quadrants through the same ordered matrix that dithers images, so it CLAIMS half – collision stays honest and the result survives into a font as real contours.
opacity changes how the SAME geometry is painted; the element still claims every quadrant it did at full strength, which is what L019 exists to catch. They are separate controls on purpose. Do not reach for opacity to make an overlap go away.

pattern: "dashed" | "dotted" rhythm ALONG the path

A dash is keyed to DISTANCE TRAVELLED, not to the lattice, so it keeps its rhythm around a corner and reads as one broken line. Keying it to the lattice the way tone is keyed would restart the cycle at every turn and produce a line that looks damaged rather than dashed. Use it for a projected trendline, a leader, or any boundary the reader should understand as inferred.

The threshold keys off absolute lattice position, so two toned shapes tile seamlessly where they meet and the same command always inks the same quadrants. Below 0.0625 nothing inks at all, so it is rejected rather than drawn as an invisible element that still occupies space.

DRAWING FROM A SOURCE – reach for this BEFORE deriving geometry by hand

measure_image source [maxWidthCells|maxHeightCells] [fit]
Reports the measured whole-cell footprint plus separate embedded-pixel and dither/simplify quadrant scales. Read UPSCALE, DOWNSCALE, or EXACT before placement.
Upscaling repeats/interpolates existing information; it creates no detail.
place_image id at span source [mode] [fit] [detail] [supersample] [page]
mode "simplify" intentionally makes a perceptual approximation rather than a 1:1 copy. Near-binary sources use clean contrast thresholding; continuous-tone sources use colour-aware contour selection and raise L023 because geometry cannot know the subject. detail is auto|low|medium|high. Fewer than 24 quadrants on the short side is refused; use a larger span or purpose-built icon artwork. supersample is auto|1|2|4. A factor of 4 builds a working canvas at 4x width and height, then box-averages each 16-sample block into one of 17 exact final coverage levels. Runs retain that coverage through save/reopen. auto prefers 4x within limits.
mode "dither" quantises the image ONTO the lattice through a 4x4 Bayer matrix. Real quadrants, merged into runs, byte-identical every run. Downscale area-averages; upscale repeats nearest samples. PNG decodes on node:zlib alone. L022 blocks a busy checker result. Remove and re-place to change its span.
mode "embed" (default) keeps a picture as a picture; it still claims an exact footprint and collides like any other element. Use it for photos and evidence. Resize recomputes its scale report.
fit "contain" (default) preserves every edge with possible padding;
"cover" fills the footprint by cropping overflow. Both preserve aspect.
place_reference source span [at] [opacity] [id] [mode] [fit] [detail] [supersample]
Lays a dithered or simplified copy UNDER the drawing to trace over, flagged L020 until remove_page takes it out, so scaffolding cannot ship.

If a shape has to LOOK like something real – a brain, a leaf, a face – a formula will not get there. Sine waves are not cortex. Simplify prepared source art, dither it, or trace a reference. Hand-computing a bitmap is the long way round, and the usual reason an author reaches for it is that they did not know these exist.

dir	up down left right	n counts whole 10px cells
align	vertical strokes: left right	horizontal strokes: top bottom (no centre – a 5px stroke centred in a 10px cell starts at 2.5px)
style	square rounded indented chamfered	
sides	top bottom left right – a corner names the two sides it connects;	one of them must be the side the path arrives on

EXAMPLE – a connector between two boxes, with no address arithmetic

```
pen from gateway.S
down align right line to checkout.N arrow
```

CONNECTORS: THE TWO MISTAKES WORTH KNOWING

1. Starting at an address you worked out yourself. A box's south face is already outside it, but its north face is its own top row – so leaving northward starts one quadrant higher. Use "pen from <id>.<face>" instead; place_box and describe also report the seat address for every face.
2. Assuming "to <id>.<port>" arrives. It only sets the DISTANCE along the way you are travelling. If the run is on a different row or column from the target, it stops level with it and never touches it – reported as L016.

EVERY FIX HAS A TOOL

widen / heighten	-> resize	shorten / font	-> restyle
move	-> move	rename	-> rename
intent	-> update_page	canvas	-> set_canvas
extend	-> extend_path	reroute	-> replace_path
offset / hop	-> replace_path with a hop piece or the other alignment		

```
lattice:
{
  "pxPerCell": 10,
  "pxPerQuadrant": 5,
  "quadrantsPerCell": 2,
  "strokeWidthPx": 5,
  "addressing": "Excel: columns A..Z, AA.., rows 1.. ; origin A1 top-left; unbounded right and down",
  "precisions": {
    "cell": "C4",
    "pin": "C4.tl (9 per cell)",
    "quadrant": "C4.q2 (4 per cell)"
  },
  "pins": [
    "tl",
    "t",
    "tr",
    "l",
    "c",
    "r",
    "bl",
    "b",
    "br"
  ],
  "font": {
    "size": 10,
    "advancePx": 6,
    "lineHeightPx": 15,
    "paddingQuads": 1
  },
  "capacity": {
    "charsPerCellWidth": 1.6666666666666667,
    "formula": "chars per line = floor((cellsWide * 10 - 10) / 6)"
  },
  "strokeAlignments": {
    "vertical": [
      "left",
```

```
      "right"
    ],
    "horizontal": [
      "top",
      "bottom"
    ],
    "note": "no centre: a 5px stroke centred in a 10px cell would start at 2.5px, off the lattice"
  },
  "cornerStyles": [
    "square",
    "rounded",
    "indented",
    "chamfered"
  ],
  "legibilityFloorPx": 8
}
```

```
rules:
{
  "L001": {
    "severity": "S0",
    "title": "node overlap",
    "blurb": "two elements claim the same quadrants on one page"
  },
  "L002": {
    "severity": "S1",
    "title": "label too wide",
    "blurb": "measured text is wider than the box interior"
  },
  "L003": {
    "severity": "S1",
    "title": "label too tall",
    "blurb": "wrapped text needs more lines than the box can show"
  },
  "L004": {
    "severity": "S1",
    "title": "stroke crosses node",
    "blurb": "a path runs through the inked body of a box"
  },
  "L005": {
    "severity": "S1",
    "title": "exclusive page overlap",
    "blurb": "a page declared exclusive overlaps content below it"
  },
  "L006": {
    "severity": "S2",
    "title": "stroke overlap",
    "blurb": "two paths claim the same quadrant with no junction"
  },
  "L007": {
    "severity": "S2",
    "title": "no gutter",
    "blurb": "two boxes touch with no separating quadrant"
  },
  "L008": {
```

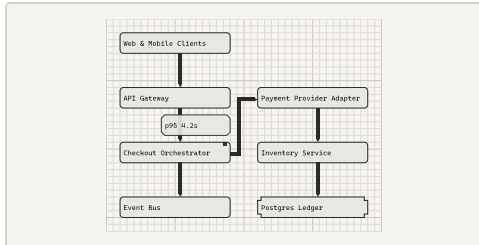
```
"severity": "S2",
"title": "dangling path end",
"blurb": "a path ends without meeting a box or another path"
},
"L009": {
  "severity": "S2",
  "title": "below legibility floor",
  "blurb": "font size under 8px"
},
"L010": {
  "severity": "S3",
  "title": "overlay overlap",
  "blurb": "expected overlap from a page declared as an overlay"
},
"L011": {
  "severity": "S2",
  "title": "outside canvas",
  "blurb": "an element extends past the declared canvas bounds"
},
"L012": {
  "severity": "S0",
  "title": "duplicate id",
  "blurb": "the same element id appears more than once"
},
"L013": {
  "severity": "S3",
  "title": "passes through corner cut",
  "blurb": "a path crosses a claimed but un-inked corner quadrant"
},
"L014": {
  "severity": "S2",
  "title": "path discontinuity",
  "blurb": "a stroke was drawn on a track the cursor was not on"
},
"L015": {
  "severity": "S2",
  "title": "path self-overlap",
  "blurb": "a path re-draws a quadrant it already covered"
},
"L016": {
  "severity": "S2",
  "title": "target not reached",
  "blurb": "a path named a destination but stops short of touching it"
```

```
},
"L017": {
  "severity": "S3",
  "title": "centring bias",
  "blurb": "centred text could not be split evenly, so a pixel went left"
},
"L018": {
  "severity": "S3",
  "title": "centring bias",
  "blurb": "a stroke centred in an even corridor could not sit exactly in the middle"
},
"L019": {
  "severity": "S2",
  "title": "invisible but claiming",
  "blurb": "an element faded past legibility still occupies its quadrants"
},
"L020": {
  "severity": "S2",
  "title": "reference still present",
  "blurb": "a tracing underlay is still in the document"
},
"L021": {
  "severity": "S1",
  "title": "overlay obscures text",
  "blurb": "opaque overlay content crosses a lower text run"
},
"L022": {
  "severity": "S2",
  "title": "busy raster image",
  "blurb": "high-frequency black/white transitions obscure image identity"
},
"L023": {
  "severity": "S2",
  "title": "heuristic image approximation",
  "blurb": "continuous-tone source was simplified without semantic understanding"
},
"C001": {
  "severity": "S3",
  "title": "sparse canvas",
  "blurb": "the page has so little ink that nothing was really composed"
}
}
```

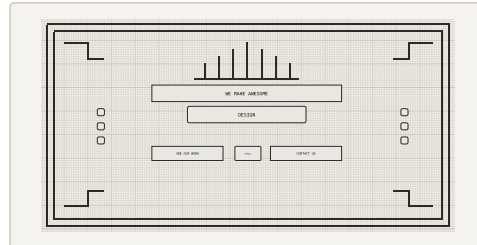
EVERYTHING TURTLEPEN HAS DRAWN

33 SVGS

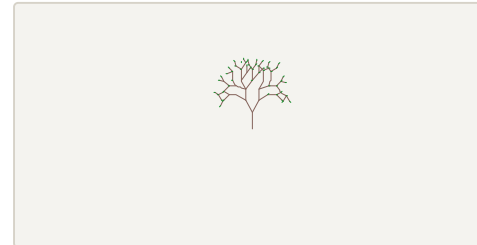
Every SVG in the project. These are the corpus the composition thresholds are calibrated against — which is why **C001** cannot simply be loosened to suit whatever diagram is in front of you.



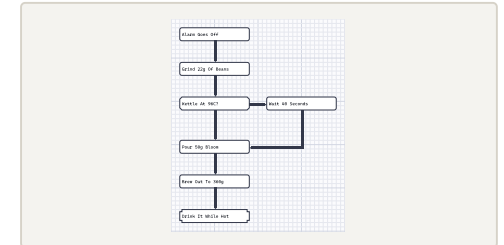
agent-session.svg
verification run: an agent authoring over real MCP
14 KB



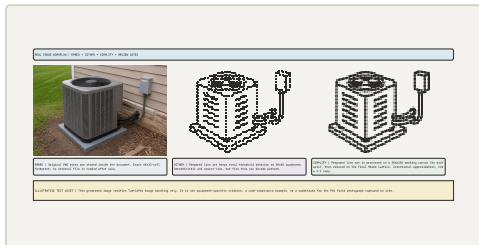
art-deco-hero.svg
dense decorative composition study
157 KB



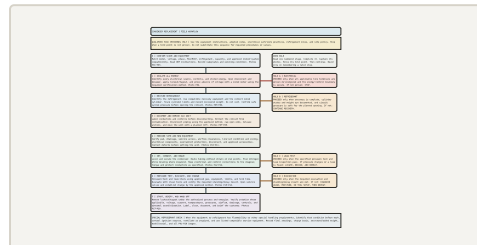
branching-tree.svg
65 branch segments + 50 leaves on an overlay page
13 KB



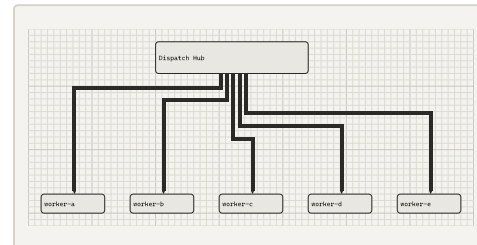
coffee-ritual.svg
narrative sequence study
15 KB



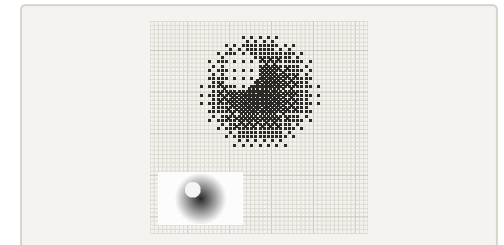
condenser-image-workflow.svg
photo embed / dither / simplify proof (4.5 MB)
4,479 KB



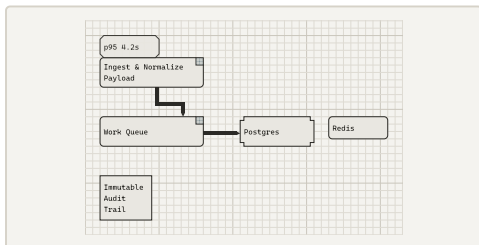
condenser-replacement-field-guide.svg
HVAC field guide, 27 committed operations
18 KB



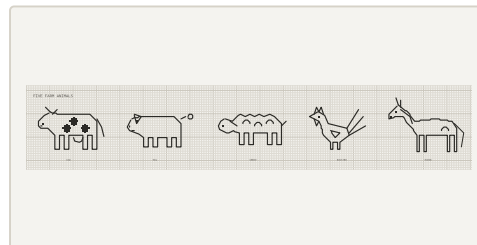
constraint-stress.svg
crowded-route constraint stress example
30 KB



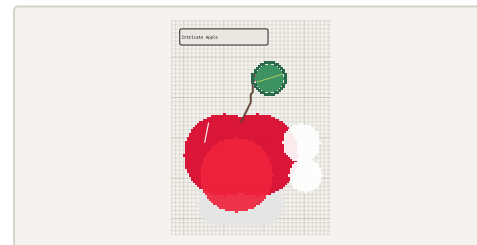
dither-proof.svg
Bayer dither determinism proof
51 KB



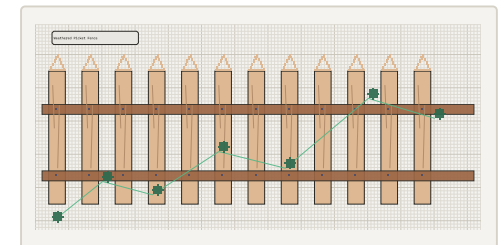
example.svg
the canonical two-box connector example
12 KB



farm-animals.svg
this document - five silhouettes, legs folded into the outline
27 KB



gemini31-scene-apple.svg
Gemini 3.1 issue-log rebuild
23 KB

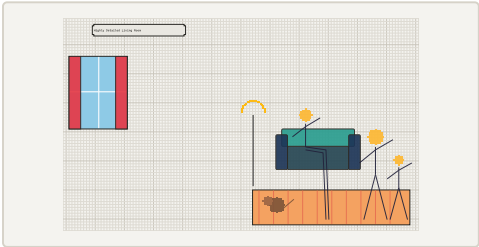


gemini31-scene-fence.svg
Gemini 3.1 issue-log rebuild
39 KB

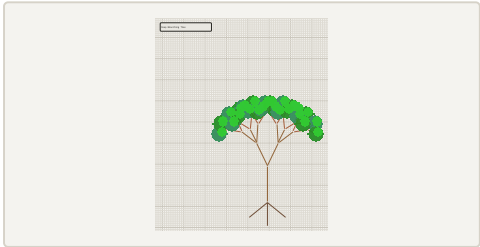
EVERYTHING TURTLEPEN HAS DRAWN

33 SVGS

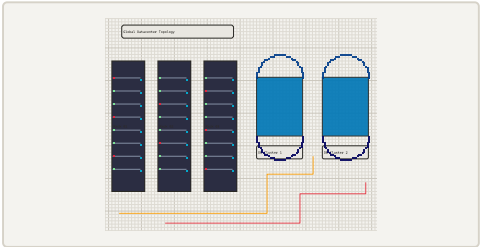
CONTINUED



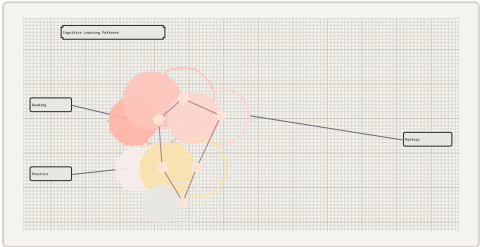
gemini31-scene-living-room-family.svg
Gemini 3.1 issue-log rebuild
24 KB



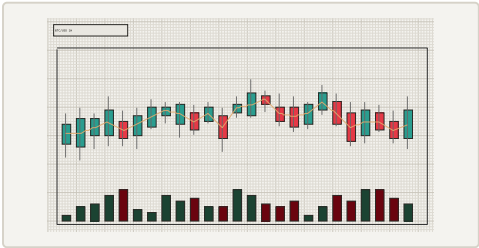
gemini31-scene-tree.svg
Gemini 3.1 issue-log rebuild
59 KB



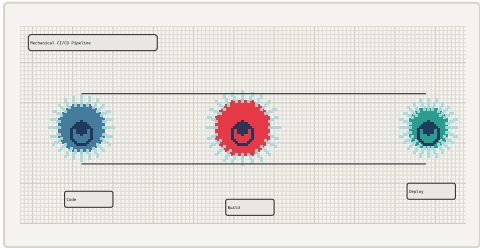
gemini31-server-structure.svg
Gemini 3.1 issue-log rebuild
31 KB



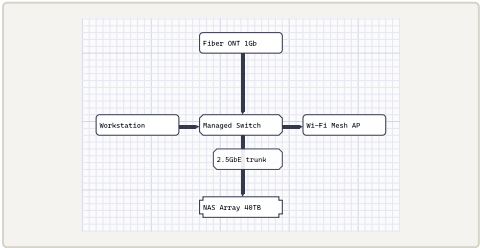
gemini31-teaching-loop.svg
Gemini 3.1 issue-log rebuild
42 KB



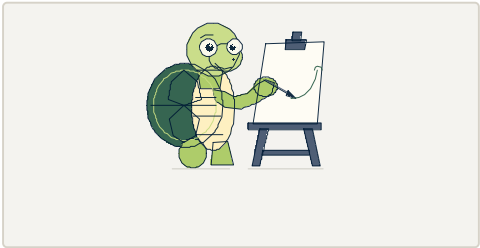
gemini31-technical-analysis.svg
Gemini 3.1 issue-log rebuild
24 KB



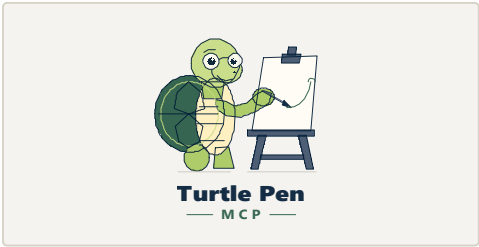
gemini31-workflow.svg
Gemini 3.1 issue-log rebuild
45 KB



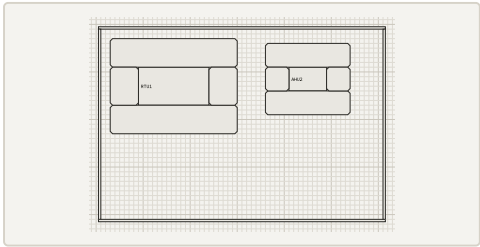
home-lab-network.svg
network diagram - 100% straight runs,
legitimately
10 KB



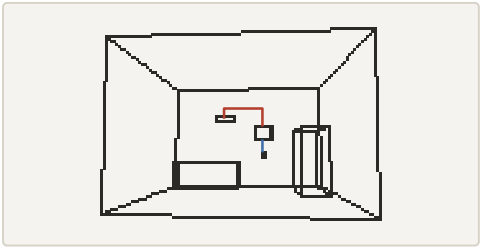
logo-mark.svg
logo mark alone
54 KB



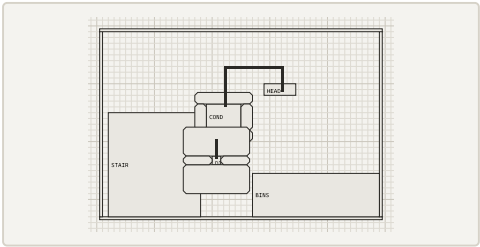
Logo.svg
canonical brand logo, authored on the lattice
54 KB



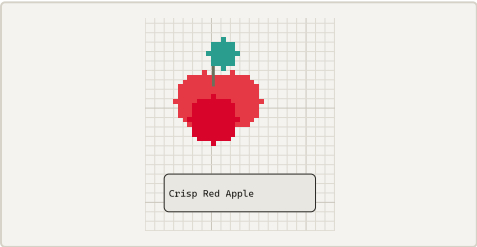
mechroom.svg
mechanical room plan
10 KB



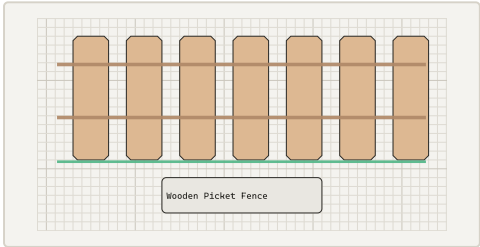
minisplit-3d.svg
perspective_scene projection from real room
inches
66 KB



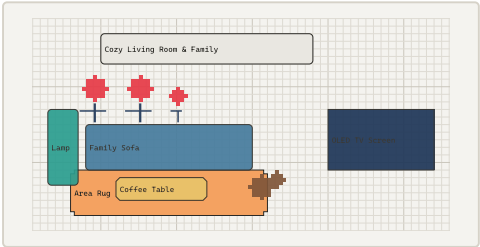
minisplit-elevation.svg
dimensioned wireframe elevation
12 KB



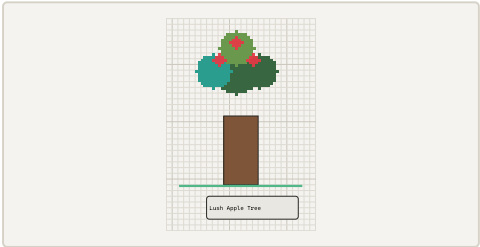
scene-apple.svg
scene study
8 KB



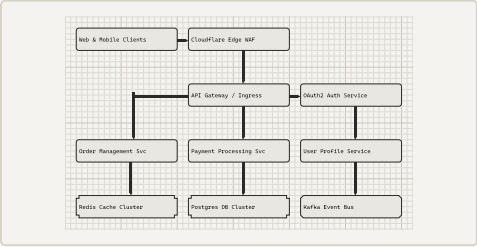
scene-fence.svg
scene study
8 KB



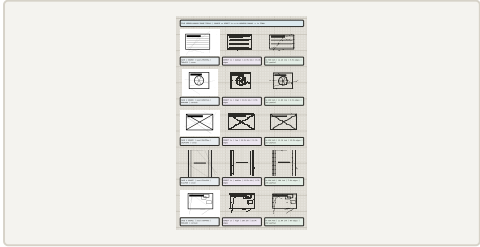
scene-living-room-family.svg
scene study
12 KB



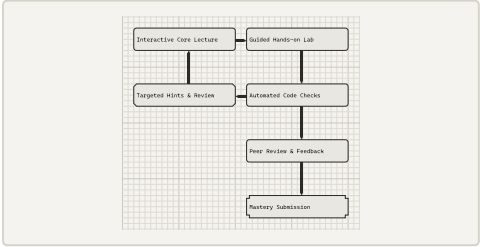
scene-tree.svg
scene study
11 KB



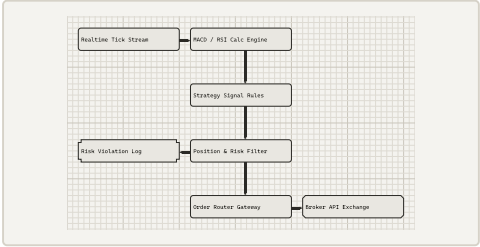
server-structure-ha-microservices.svg
HA microservices architecture
18 KB



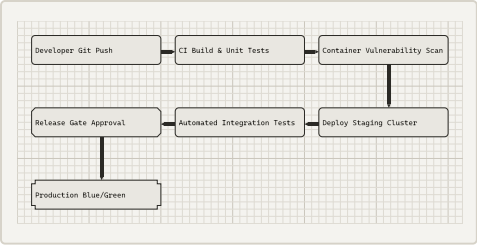
supersample-random-five.svg
five seeded-random 4x supersample trials
185 KB



teaching-mastery-learning-cycle.svg
mastery learning cycle
12 KB



technical-analysis-quant-engine.svg
quant engine technical analysis
13 KB



workflow-cicd-deployment-pipeline.svg
CI/CD deployment pipeline
12 KB